

RANGE-ADAPTIVE DOUBLE-PRECISION GEMM EMULATION ON INTEGER TENSOR CORES WITH EXACT RESIDUE RECONSTRUCTION

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ABSTRACT. Most consumer and workstation GPUs execute double-precision arithmetic at one sixty-fourth of their single-precision rate, while their INT8 tensor cores are the fastest units on the die. We present RARE, range-adaptive residue emulation, implemented in the open library `fp64-emu`: double-precision matrix multiplication executed as one INT8 GEMM per pairwise-coprime modulus and recovered exactly through the Chinese Remainder Theorem, following the Ozaki-II scheme. A runtime Cauchy–Schwarz pass measures the operands and converts unused residue range into operand precision, making every modulus count range-safe; extraction runs entirely in integer arithmetic with round-to-nearest splitting, and two additional INT8 GEMMs cancel the first-order quantization error; reconstruction computes balanced mixed-radix digits with packed byte dot products held in registers; and a fused tensor-core engine reduces every product to a one-byte residue in its epilogue, dispatched against cuBLAS by measured per-shape timing. We prove range safety, bit-exactness on representable inputs, and a worst-case error bound that quantifies the correction. Measured on five NVIDIA architectures, `fp64-emu` exceeds native FP64 by up to a factor of 26 on the FP64-throttled parts and exceeds `GEMMu18`, the scheme authors’ reference implementation, at every size on every such part at equal GEMM counts, by 1.07 to 3.80 times; on unthrottled datacenter silicon, emulation pays only at scale, and there `GEMMu18` leads. A tiled Cholesky factorization routing its trailing updates through the batched entry point runs 1.5 times faster end to end at equal residuals. Accuracy exceeds native DGEMM in every distribution tested, with exactness verified against integer oracles to depth 200,000.

1. INTRODUCTION

The arithmetic that modern GPUs are best at is not the arithmetic that scientific computing asks for. A current consumer or workstation GPU executes IEEE double precision at 1/64 of its single-precision rate: an RTX 6000 Ada sustains roughly 1.3 TFLOP/s in FP64 against hundreds of tera-operations per second on its INT8 tensor cores, and the recent Blackwell workstation parts widen this ratio further. Deep learning workloads dominate the economics of GPU design, and full-rate double-precision units survive only on a shrinking set of datacenter parts. For the large community computing on everything else, double-precision matrix multiplication is effectively a legacy path executed at a small fraction of the machine’s capability.

Emulation inverts this situation. If a double-precision product can be decomposed into a modest number of low-precision matrix multiplications, the idle tensor cores pay for the decomposition many times over. The question is what the decomposition costs in accuracy, generality, and overhead, and the answer has improved in three generations of work. The Ozaki scheme [1] splits each operand into high-precision slices and forms the product as a sum of error-free partial products; mapped onto integer tensor cores by Ootomo, Ozaki, and Yokota [3] and refined by Uchino, Ozaki, and Imamura [4], it emulates DGEMM with a number of low-precision GEMMs that grows quadratically in the slice count. The Ozaki-II scheme of Ozaki, Uchino, and Imamura [5] replaces slicing with residue arithmetic: the scaled integer operands are reduced modulo a set of small pairwise-coprime moduli, one INT8 GEMM computes the product in each residue class, and the Chinese Remainder

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Theorem recovers the exact integer product. The modulus count, and with it the GEMM count, is now linear in the recovered precision. The authors’ open-source library, **GEMMu18** [6], is the reference implementation of this scheme and the measured state of the art.

This paper presents RARE, a range-adaptive residue emulation of double-precision GEMM implemented in the **fp64-emu** library. On the architectures whose FP64 units are throttled, which are the parts this class of method exists for, it is faster than **GEMMu18** at every size at equal GEMM counts, while extending the scheme’s guarantees in several directions. The contributions are as follows.

Runtime operand sizing (Section 3.1). Ozaki-II sizes its integerization from a worst-case bound: the user fixes the modulus count, and the scaling width k is derived so that even K fully aligned terms at full magnitude cannot overflow the residue range. Real data sits far below this worst case. **fp64-emu** measures the operands at run time, in one pass fused with the scaling pass, and applies the Cauchy–Schwarz inequality to bound the actual product: whatever residue range the data leaves unused is converted into additional operand mantissa bits, and when the data needs more range than the configured moduli provide, the operands are narrowed instead. The consequence is a scheme in which every configuration the library plans is safe. Adversarial input costs correct bits; it cannot corrupt the answer.

Integer-only extraction with first-order correction (Section 3.2). The pass that converts scaled doubles into balanced residues is memory-bound in principle but was compute-limited in practice: on parts whose FP64 pipe runs at 1/64 rate, even the handful of floating-point operations in a rounding-and-scaling step co-limits the kernel. Our extraction operates on the raw IEEE encoding in integer registers throughout, performing round-to-nearest-even by explicit shift arithmetic and reducing each scaled integer to all residues through a three-limb 32-bit Barrett scheme. Where Ozaki-II truncates toward zero and controls the resulting error through its worst-case width, we round to nearest and then cancel the first-order quantization error explicitly: two additional INT8 GEMMs over eight-bit summaries of the operands and their residuals recover the cross terms, lifting accuracy past the ceiling that operand quantization alone imposes and making the default configuration more accurate than native DGEMM.

Register-resident balanced-Garner reconstruction (Section 3.3). Ozaki-II reconstructs through the classical CRT sum, accumulating $C_i \cdot M_i \cdot y_i$ in wide arithmetic and reducing modulo the full modulus product. We instead compute balanced mixed-radix digits by Garner’s recurrence, reformulated so that the digit inner products run as packed byte-dot-product (**dp4a**) instructions over digits held four to a register, and the final recombination collapses to one 64-bit Horner step per group of seven moduli. The kernel keeps every intermediate in registers; on the architecture where this pass was compute-bound, the reformulation made it $1.6\times$ faster, and the exact path produces the correctly rounded double of the reconstructed integer.

A fused residue engine with measured dispatch (Section 4). **GEMMu18** executes its modular products through cuBLAS INT8 GEMMs into full 32-bit product planes that a separate pass reduces. **fp64-emu** carries an in-file tensor-core kernel whose epilogue Barrett-reduces every product to its balanced residue and stores one byte per element, quartering the intermediate traffic; its operands are staged through a tile-blocked layout written by the extraction pass in exactly the order the kernel consumes, and on Hopper and Blackwell the staging uses the Tensor Memory Accelerator. Because neither engine wins everywhere, the dispatcher times both once per problem shape and caches the choice, a policy that remains correct on architectures we never measured. The same infrastructure supports batched products through plane-batch slots, consumption of the standard $[K, N]$ operand layout without a transpose copy, products of depth up to $K = 8.4$ million through segmented residue accumulation, and capture into CUDA graphs, under which the emulation outpaces native FP64 from matrix dimension 64 upward.

Measurement (Section 5). We evaluate on five NVIDIA architectures: GeForce RTX 3070 Ti (consumer Ampere, SM 8.6), RTX 6000 Ada (SM 8.9), RTX PRO 6000 Blackwell Server Edition

(SM 12.0), A100-SXM4-80GB (datacenter Ampere, SM 8.0), and H200 (Hopper, SM 9.0). Against native FP64, the default configuration reaches $26.1\times$ at $n = 8192$ on Blackwell at accuracy above native DGEMM in every distribution and depth combination tested. Against `GEMMu18`, built from source and measured on the same hardware with the GEMM count held equal, `fp64-emu` is faster at every size on the three throttled parts, by $1.07\times$ to $3.80\times$; on Hopper, whose FP64 is unthrottled, `GEMMu18` leads at the largest sizes. On exactly representable inputs the default reproduces the true product bit for bit, verified by equality against integer oracles including an adversarial fully aligned case and a dyadic dynamic-range case.

The implementation is a single CUDA source file with a Python surface, published with all measurement scripts [16].

2. BACKGROUND AND RELATED WORK

2.1. The Ozaki line. The original Ozaki scheme [1] is an error-free transformation of matrix multiplication: each operand is split into slices whose products are exactly representable, so a high-precision product becomes a sum of low-precision GEMMs. Its appeal on modern hardware is that the low-precision GEMMs can be executed by matrix engines; Mukunoki, Ozaki, Ogita, and Imamura demonstrated DGEMM on FP16 tensor cores this way [2], and Ootomo, Ozaki, and Yokota moved the scheme to integer units [3], where the multiply-accumulate is exact and the slice products are error-free by construction. Uchino, Ozaki, and Imamura [4] improved the integer variant’s accuracy and throughput. The cost structure of all slicing schemes is quadratic: s slices per operand require $s(s + 1)/2$ GEMMs.

Ozaki-II [5] abandons slicing for residue arithmetic. Operands are scaled by powers of two and truncated to integers; each integer matrix is reduced modulo s pairwise-coprime moduli chosen so that balanced residues fit INT8; one INT8 GEMM per modulus computes the product in each residue class, exactly, because a sum of products of values bounded by 128 in magnitude stays within INT32 for any practical inner dimension; and the true integer product is recovered from its residues by the Chinese Remainder Theorem, provided it lies within the signed range of the modulus product. The GEMM count is now linear in the recovered precision: roughly eight bits per modulus. The reference implementation `GEMMu18` executes the modular GEMMs through `cublasGemmEx` and exposes the modulus count and a fast/accurate mode pair as tuning parameters; its fast mode estimates the output magnitude by the Cauchy–Schwarz inequality, while its accurate mode spends one additional INT8 GEMM to compute the bound exactly and reduce the scaling overestimate. We adopt from `GEMMu18` its modulus table, whose composite entries ($255 = 3 \cdot 5 \cdot 17$, 253, 247, 217) carry nearly eight bits each where primes alone would carry fewer; balanced residues make the ceiling 256 rather than 128, since for even m the balanced representatives lie in $[-m/2, m/2 - 1]$, which at $m = 256$ is exactly the signed eight-bit range $[-128, 127]$.

2.2. Other emulation paths. NVIDIA has announced floating-point emulation inside cuBLAS itself [10], currently centered on FP32 emulation via BF16 tensor cores, with FP64 support on datacenter parts; the numerical behavior of tensor-core arithmetic that such schemes build on is analyzed in [7, 8]. Mixed-precision iterative refinement [9] attacks the same economics from the solver side, running factorizations in low precision and recovering accuracy through residual iteration; emulated DGEMM is complementary, serving the many workloads that want a drop-in high-precision product rather than a restructured solver. Since the Ozaki-II preprint, an FP8 variant [11] and fused-kernel treatments [12] have appeared, and the scheme has been evaluated inside factorization workloads [13]; these confirm an active design space in which the measured comparison of Section 5 situates our contribution.

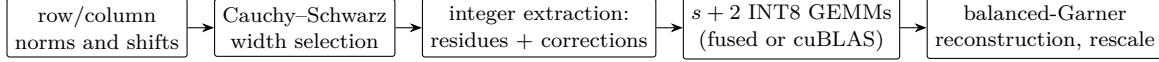


FIGURE 1. The **fp64-emu** pipeline. All five phases execute on the GPU; the sizing decision is made on device, so no synchronization with the host occurs between phases.

3. THE PIPELINE

Figure 1 shows the five phases of RARE as implemented in **fp64-emu**. This section walks through the numerically substantive ones: the sizing pass that chooses the operand width, the extraction that produces residues and correction operands, and the reconstruction. Section 4 covers the engine that executes the modular GEMMs and the dispatch machinery around it.

3.1. Scaling and runtime operand sizing. Both the slicing and residue schemes begin the same way: each row of A and each column of B is scaled by a power of two so that its largest element becomes an integer carrying the leading mantissa bits. Writing $\tilde{A} = \text{round}(A 2^{\sigma_i})$ row-wise with $|\tilde{A}_{ij}| < 2^b$, the exact integer product $P = \tilde{A}\tilde{B}$ determines $C = AB$ up to the per-row and per-column power-of-two rescale, and the whole numerical question becomes whether P fits inside the signed range $M/2$ of the modulus product $M = \prod_i m_i$ and how much of each operand’s mantissa the width b retains.

The tension is that these two quantities trade against each other through a bound that is almost never tight. In the worst case, an inner product of K terms, each the product of two b -bit integers of the same sign at full magnitude, requires $2b + \lceil \log_2 K \rceil + 1$ bits. Ozaki-II provisions for exactly this case: given the user’s modulus count, it derives the width from the worst-case rule, which for random data wastes most of the provisioned range, since a random inner product grows like $\sqrt{K}$ rather than K .

fp64-emu measures instead. The scaling pass, which must in any case scan every element once to find each row’s maximum, additionally accumulates each row’s 2-norm. By the Cauchy-Schwarz inequality,

$$|P_{ij}| = \left| \sum_k \tilde{A}_{ik} \tilde{B}_{kj} \right| \leq \|\tilde{A}_{i,:}\|_2 \|\tilde{B}_{:,j}\|_2 \leq \max_i \|\tilde{A}_{i,:}\|_2 \cdot \max_j \|\tilde{B}_{:,j}\|_2,$$

a rigorous bound computable from two scalar reductions. A second, tiny kernel compares this bound against the configured range $\log_2 M$ and chooses the largest additional shift g such that widening both operands by g bits, which scales the product bound by 2^{2g} , still fits with half a bit of margin; the margin dominates both the rounding slack of the scaled integers and the accumulation error of the norm reduction. The shift folds into the per-row scales, and the rest of the pipeline sees a single effective width $b_{\text{eff}} = b + g$.

Two consequences follow. First, on typical data the default configuration recovers the full 53-bit significand with fewer moduli than the worst-case rule would demand: for dense Gaussian rows the norm-product bound scales as $2b + \log_2 K - \log_2(2 \ln K)$, about $2b + 8$ at $K = 4096$ against the worst-case $2b + \log_2 K$, and for rows dominated by a few large entries the saving approaches the full $\log_2 K$; Section 5.4 quantifies what the pass delivers. Second, and more important for robustness, g may be negative: if the caller provisions fewer moduli than the data requires, the operands are narrowed until the bound fits. Narrowing is bounded below by the width floor $b_e \geq 16$ discussed with Proposition 3.1; every modulus count above that floor’s envelope, including every count the planner emits, is range-safe, and the count becomes a pure accuracy-versus-speed knob. **GEMMv18**’s fast mode also invokes Cauchy-Schwarz, but for a different purpose, estimating the output magnitude to set its scaling constant against a fixed width; there, an overestimate costs accuracy. Here the inequality allocates the width itself, and an overestimate merely leaves a fraction

of a bit unclaimed. We additionally cap the usable range at 2^{126} so that the 128-bit reconstruction path of Section 3.3 remains sound at any depth; the cap is invisible below twenty moduli.

The bound degrades gracefully on adversarial input. For fully aligned data, all terms of one sign at full magnitude, the norms detect exactly the worst case and the operands narrow accordingly; the test suite includes an all-ones product at $K = 2048$, reproduced exactly. Non-finite data drives the norm to infinity, which disables widening and falls back to the worst-case rule.

3.2. Integer-only extraction and the correction planes. The extraction pass reads each scaled double once and writes s residue planes plus two correction operands. It is, by construction, a streaming pass whose floor is memory bandwidth, and reaching that floor required removing every FP64 instruction from it: on SM 8.6, profiling attributed a co-limiting share of the pass to the 1/64-rate FP64 pipe through the `nearbyint/ldexp` formulation. The shipped kernel therefore operates on the raw IEEE encoding in integer registers. The power-of-two scale is an exponent addition; round-to-nearest-even of $\text{mag} \cdot 2^{-t}$ is an explicit shift with tie handling on the retained low bit; and the result, biased by $2^{b_{\text{eff}}}$ to make it non-negative, splits into three 18-bit limbs from which each balanced residue follows by

$$r = (\ell_2 c_{36} + \ell_1 c_{18} + \ell_0 + c_b) \bmod m, \quad c_{36} = 2^{36} \bmod m, \quad c_{18} = 2^{18} \bmod m,$$

with c_b chosen to cancel the bias inside the sum. Each limb product stays below 2^{26} and the limb sum below 2^{27} , so the Barrett quotient underestimates the true quotient by at most one and a single conditional subtraction completes the reduction; no 64-bit multiply-high appears anywhere in the pass. The integer formulation is also exact where an underflowing `ldexp` would round. Measured in isolation, the pass runs at 92% of DRAM bandwidth on SM 8.6.

Rounding to nearest halves the worst-case quantization residual relative to Ozaki-II's truncation, but the more consequential departure is that we then cancel the first-order error explicitly. Write $\tilde{A} + r_A = A 2^\sigma$ with $|r_A| \leq \frac{1}{2}$ element-wise. The exact product decomposes as

$$A 2^\sigma B 2^\tau = \tilde{A} \tilde{B} + (\tilde{A} r_B + r_A \tilde{B}) + r_A r_B,$$

and the middle term, the first-order quantization error, is itself a pair of matrix products over small operands. During extraction we emit, per element, an eight-bit summary $h = \text{round}(\tilde{A} 2^{7-b_{\text{eff}}})$ of the integer's leading bits and an eight-bit quantization residual $q = \text{round}(r \cdot 2^8)$, both spanning the full signed INT8 range. Two additional INT8 GEMMs, $h_A q_B^\top$ and $q_A h_B^\top$, then recover the first-order term at scale $2^{b_{\text{eff}}-15}$, and the reconstruction folds it in before the single final rounding. On integer inputs the residuals vanish identically and the correction is exact silence. On general inputs the correction lifts accuracy roughly four bits past the quantization ceiling, which is what carries the default configuration above native DGEMM (Section 5.5); it costs two GEMMs against the roughly fourteen the default already runs. Ozaki-II has no analogous term; its accurate mode spends its extra GEMM on a tighter magnitude bound rather than on error cancellation.

3.3. Reconstruction. After the modular GEMMs, each output element exists as s balanced residues, and the remaining work is to recover the integer they encode and scale it into the output double. This pass is elementwise: at $s \approx 14$ moduli it reads fourteen bytes of residues and eight of correction data per element while performing an $O(s^2)$ digit recurrence, a profile that is neither bandwidth-bound nor negligible.

We reconstruct by Garner's mixed-radix recurrence rather than by the classical CRT sum that Ozaki-II uses. The mixed-radix digits

$$d_t = \left(r_t - \sum_{u < t} d_u (M_u \bmod m_t) \right) (M_t \bmod m_t)^{-1} \bmod m_t, \quad M_u = \prod_{v < u} m_v,$$

computed in the balanced representative range $[-m_t/2, m_t/2)$, fit signed bytes, and every factor $M_u \bmod m_t$ can be re-balanced to a byte as well without leaving its congruence class. The inner

product of the recurrence then becomes a dot product of two byte-vectors, which GPUs execute four lanes at a time by the `dp4a` instruction. We pack the digits four to a 32-bit register as they are produced and pre-pack the factor table with zeros above the diagonal, so each digit costs a fixed number of `dp4a` operations, one 32-bit Barrett reduction, and no tail handling: at nineteen prior digits the whole recurrence step stays below 2^{19} and never needs 64-bit arithmetic.

The final recombination $P = \sum_t d_t M_t$ is a Horner evaluation whose naive form costs one 128-bit multiply-add per modulus, and profiling on SM 8.6 showed the kernel bound on the pipe that executes wide integer multiplies. We collapse the evaluation into groups: a Horner pass over at most seven moduli stays below 2^{55} and runs in plain 64-bit registers, and the groups combine high to low with a single wide multiply each, in the three-group form $P = a_0 + M_{0..6} (a_1 + M_{7..13} a_2)$. At the default fourteen moduli two groups suffice, so one wide multiply replaces thirteen 128-bit steps; counts above fourteen engage the third group and a second multiply. A register-residency constraint completes the design: both digit loops unroll to the compile-time modulus maximum and exit at the runtime count, because any runtime-indexed thread-local array is demoted to local memory, which we measured as a $2.4\times$ amplification of the kernel’s DRAM traffic. The grouped, register-resident form made the reconstruction $1.6\times$ faster on SM 8.6 and $1.4\times$ faster on SM 8.9, and left it at its measured traffic floor on the latter.

The exact path performs the recombination in 128-bit integer arithmetic carried as two 64-bit halves, adds the correction term at its scale, and converts once to double with correct rounding, folding the dropped tail into a sticky bit far below the round position. The result is the correctly rounded double of the corrected integer product. A fast path replaces the 128-bit combine with one in double precision, trading the last bit of the accumulator for a few percent of the phase; both paths keep the recombination off the throttled native-FP64 pipe.

3.4. Guarantees. The pipeline’s numerical claims reduce to three statements: the sizing pass never admits a product that wraps the residue range, the arithmetic between extraction and output is exact, and the correction bounds the error on general inputs. We state and prove them here; throughout, b_e denotes the effective operand width, σ_i and τ_j the per-row and per-column shifts after widening, $\tilde{a}_{ik} = \text{round}(a_{ik} 2^{\sigma_i})$ with residual $r_{ik}^A = a_{ik} 2^{\sigma_i} - \tilde{a}_{ik}$, $|r_{ik}^A| \leq \frac{1}{2}$, and $P = \tilde{A} \tilde{B}^\top$ the exact integer product. $M^* = \min(\prod_t m_t, 2^{126})$ is the capped range.

Proposition 3.1 (Range safety). *Let $b_e \geq 16$ and $K \leq 2^{23}$, and suppose the widening pass accepts the shift under its half-bit margin, that is, $\lambda_A + \lambda_B + \frac{1}{2} \leq \log_2 M^* - 1$, where λ_A is the computed value of $\max_i \log_2 \|a_{i,:} 2^{\sigma_i}\|_2$ and likewise λ_B . Then $|P_{ij}| < M^*/2$ for all i, j : the balanced CRT representative of every output element is P_{ij} itself, and the 128-bit recombination cannot overflow.*

Proof. By Cauchy–Schwarz, $|P_{ij}| \leq \|\tilde{a}_{i,:}\|_2 \|\tilde{b}_{:,j}\|_2$. Rounding perturbs each row norm by at most $\frac{\sqrt{K}}{2}$ in absolute terms, and the scaled row maximum is at least 2^{b_e-1} , so $\|\tilde{a}_{i,:}\|_2 \leq \|a_{i,:} 2^{\sigma_i}\|_2 (1 + \sqrt{K} 2^{-b_e})$, an inflation of at most $\log_2(1 + 2^{11.5-16}) < 0.064$ bits under the stated hypotheses, and likewise for columns. The norm itself is accumulated in binary64, whose relative error over at most K terms is below 2^{-30} , and stored in binary32, contributing less than 2^{-15} bits; the shifts are powers of two and perturb nothing. The total slack on the left-hand side is therefore below $2 \cdot 0.064 + 2^{-14} < \frac{1}{2}$, so the true bound satisfies $\log_2(\|\tilde{a}_{i,:}\|_2 \|\tilde{b}_{:,j}\|_2) \leq \lambda_A + \lambda_B + \frac{1}{2} \leq \log_2 M^* - 1$. $\square$

The implementation floors b_e at 16, since the correction scale 2^{b_e-15} must remain a positive shift, and does not re-check the margin once the floor binds. The hypothesis therefore has to be satisfiable at the floor: a 16-bit worst-case product needs $33 + \lceil \log_2 K \rceil$ bits of range, which seven moduli provide at every single-launch depth and eight at the segmented maximum, and the planner never emits fewer than thirteen. Only a hand-set count below that envelope can drive the floor into violation; there the hypothesis fails and the reconstruction can wrap.

Proposition 3.2 (Exactness). *Under Proposition 3.1, the exact path returns $\text{RN}(2^{-\sigma_i - \tau_j}(P_{ij} + 2^{b_e - 15}\Delta_{ij}))$, the correctly rounded double of the corrected integer value, where Δ is the integer carried by the two correction GEMMs. In particular, when every scaled entry is an integer, the residuals vanish, $\Delta = 0$, and the output is the correctly rounded true product; when that product is representable, the result is exact.*

Proof. Each stage is exact integer arithmetic. The three-limb residues are computed from the biased integer without rounding; each modular GEMM accumulates products of balanced residues bounded by 128 in magnitude, so a single launch stays within int32 for $K < 2^{17}$ and the segmented path within its int32 residue accumulators, whose entries are bounded by 128 per segment over at most 128 segments. The balanced mixed-radix digits are the unique such representation of $P_{ij} \bmod \prod m_t$, and by Proposition 3.1 that representative is P_{ij} . The grouped recombination and the correction addition are exact in 128-bit integer arithmetic within the 2^{126} cap, and the conversion folds the dropped tail into a sticky bit strictly below the round position, which preserves round-to-nearest-even. When the residuals vanish, both correction operands q are identically zero, so $\Delta = 0$ identically. The test suite witnesses the claim by bit equality against integer oracles, including the aligned worst case, a dyadic dynamic-range case, and, through the segmented path, $K = 200,000$. $\square$

Proposition 3.3 (Accuracy after correction). *Let $S_{ij} = \sum_k (a_{ik} 2^{\sigma_i})(b_{kj} 2^{\tau_j})$ be the exact scaled product and $V_{ij} = P_{ij} + 2^{b_e - 15}\Delta_{ij}$ the corrected value. With the eight-bit summaries clamped to $[-127, 127]$,*

$$|V_{ij} - S_{ij}| \leq K \left(3 \cdot 2^{b_e - 7} + \frac{9}{2} \cdot 2^{b_e - 15} + \frac{1}{4} \right),$$

and the returned double adds one rounding at the output scale. Without the correction GEMMs the corresponding bound is $K \cdot 2^{b_e}$, so the correction is worth a factor of $128/3$, about 5.4 bits, in the worst case.

Proof. Write $S - P = \sum_k (\tilde{a}r^B + r^A\tilde{b} + r^Ar^B)$ element-wise. The summaries satisfy $h = \tilde{a}2^{7-b_e} + e_h$ and $q = r2^8 + e_q$, where rounding alone gives $|e| \leq \frac{1}{2}$ and the clamp at ± 127 raises the bound to $|e| \leq \frac{3}{2}$ (the unclamped values reach ± 128). Expanding one correction pair,

$$2^{b_e - 15}h_a q_b = \tilde{a}r^B + \tilde{a}2^{-8}e_q + e_h r^B 2^{b_e - 7} + e_h e_q 2^{b_e - 15},$$

whose error terms are bounded by $\frac{3}{2} \cdot 2^{b_e - 8}$, $\frac{3}{4} \cdot 2^{b_e - 7}$, and $\frac{9}{4} \cdot 2^{b_e - 15}$ respectively, totaling $3 \cdot 2^{b_e - 8} + \frac{9}{4} \cdot 2^{b_e - 15}$ for one pair; the symmetric pair contributes the same, doubling both constants, and the uncorrected second-order term adds $|r^A r^B| \leq \frac{1}{4}$. Summing over K terms gives the stated bound. Without the correction the first-order term alone contributes up to $2 \cdot 2^{b_e - 1} = 2^{b_e}$ per term. $\square$

The bound explains the measured behavior: the corrected worst case sits about five and a half bits below the uncorrected ceiling, and the four to five bits observed on random data (Section 5.5) realize most of that margin. The clamp affects only entries whose residual magnitude exceeds $\frac{255}{512}$ and costs at most a factor of three in the dominant constant.

4. THE ENGINE AND ITS DISPATCH

4.1. A fused residue GEMM. The modular products themselves are ordinary INT8 GEMMs, and GEMMu18 delegates them to cuBLAS. Doing so fixes two costs. First, cuBLAS writes 32-bit accumulators, so $s + 2$ full INT32 product planes must be materialized and then reduced by a separate pass, a round-trip of $4(s + 2)$ bytes per output element. Second, a library GEMM consumes row-major operands at a fixed layout, which constrains the extraction pass to a write pattern that the GEMM then re-reads inefficiently.

fp64-emu carries its own tensor-core kernel for these products. The kernel computes 128×256 output tiles per thread block on `mma.sync.m16n8k32` fragments, eight warps each owning a 64×64

warp tile so that thirty-two tensor-core operations issue behind every eight fragment loads; operands stream through a four-slot shared-memory ring synchronized by per-slot arrival barriers with no block-wide synchronization in the main loop. Its distinguishing feature is the epilogue: each 32-bit accumulator is Barrett-reduced to its balanced residue inside the epilogue and stored as one byte, directly from registers, eliminating the INT32 planes and the reduction pass entirely. The correction products, which must remain 32-bit, share the same launch as two additional plane slots. On SM 9.0 and SM 12.0 the operand staging uses the Tensor Memory Accelerator with hardware swizzling; below that, asynchronous copies stage a tile-blocked operand layout that the extraction pass writes in exactly the order the kernel consumes, turning the staging into contiguous streaming reads. Ragged edges are zero-padded at extraction, so the kernel carries no bounds checks in its hot loop, and none of M , N , K carries an alignment constraint, unlike the INT8 paths of cuBLAS or `torch._int_mm`.

4.2. Measured dispatch. Neither engine dominates. The fused kernel wins wherever its byte-wide epilogue traffic and fused reduction outweigh cuBLAS’s higher raw GEMM throughput, which happens at small and medium sizes on every architecture and at all sizes on some; cuBLAS wins at the largest sizes on the parts where its INT8 kernels are strongest. The crossover moves with architecture, size, and modulus count, and any fixed rule encodes a guess that the next GPU falsifies. `fp64-emu` instead times the viable candidates once per problem shape, on the first call, warmed to exclude one-time setup, and caches the winner; the same policy chooses between cuBLAS’s strided-batched call and a per-plane loop, whose relative speed also inverts across architectures. When the INT32 planes of the cuBLAS path would not fit free memory, a chunked variant multiplies groups of planes into a bounded buffer and reduces each group to residues before the next runs. Environment overrides pin any path for measurement, and every path computes bit-identical results, which the test suite asserts.

4.3. Extensions. Three extensions widen the applicable surface without altering the numerics. *Batching*: a uniform batch flattens into plane-batch slots, so the whole batch shares one launch chain, one sizing pass with a common operand width, and one reconstruction; per-call cost is paid once, which at 64 products of dimension 256 makes the batched call 4.5× faster than looping on Blackwell and turns a regime where emulation loses to native FP64 into one where it wins (Section 5.6). *Layout*: the standard $[K, N]$ operand is consumed in place, with per-column shifts gathered by coalesced slab-reduction kernels and the residue planes written in the transposed layout that cuBLAS’s NN form consumes; the dispatcher weighs this transpose-free path against the fused engine with an internal copy, and the former removes a copy overhead of up to 14% of runtime at mid sizes. *Depth*: products beyond the single-launch bound $K < 131072$, set by the INT32 accumulator against balanced residues of magnitude 128, run in segments of 65536 whose balanced residues accumulate in INT32 planes, bounded by 128 per segment, and whose correction products accumulate in INT64; one fold and one reconstruction close the sum, preserving bit-exactness, which we verify at $K = 200,000$ against a CPU integer oracle. All entry points register fake kernels for `torch.compile` and are capturable into CUDA graphs; under graph replay, which removes the host-side launch overhead that dominates small problems, the emulated product outpaces native FP64 from $n = 64$ upward on SM 8.9.

5. EVALUATION

5.1. Setup. We measure on five NVIDIA architectures: GeForce RTX 3070 Ti Laptop (consumer Ampere, SM 8.6, 8 GB), RTX 6000 Ada (SM 8.9, 48 GB), RTX PRO 6000 Blackwell Server Edition (SM 12.0), A100-SXM4-80GB (datacenter Ampere, SM 8.0), and H200 (Hopper, SM 9.0). The first three throttle FP64 to 1/64 of the FP32 rate; the last two do not, and the results divide along exactly that line. Throughput is reported as $2n^3$ over the median of 3 to 30 timed executions after warm-up, with iteration counts scaling inversely with size; on the two locally held parts, which are

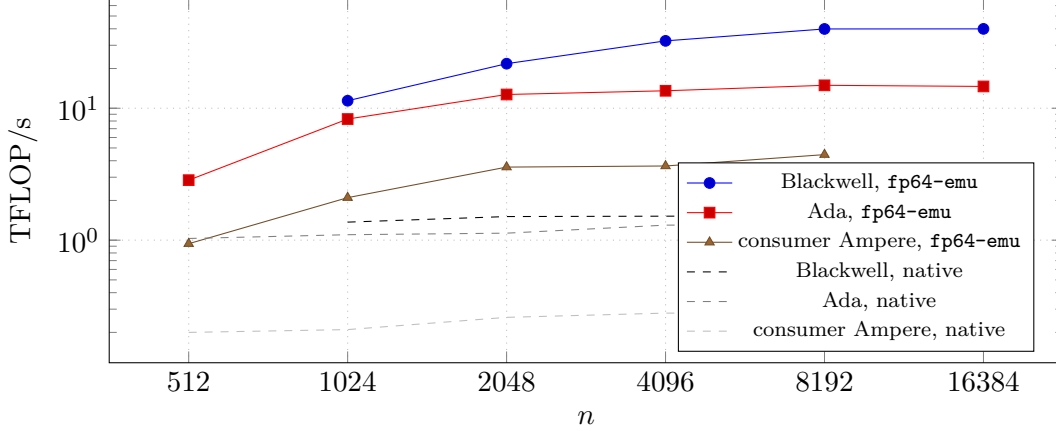


FIGURE 2. Default-configuration throughput against native FP64 on the three FP64-throttled parts. Solid: `fp64-emu`; dashed: native `torch.matmul`.

thermally variable, contending systems are additionally interleaved across repeated rounds and each cell reports its best median. The Blackwell, A100, and H200 numbers come from cloud instances. Native FP64 is `torch.matmul` on the same tensors. Software per part: consumer Ampere runs driver 580.126.09 with torch 2.7.1 on CUDA 12.6; Ada runs torch 2.10.0 on CUDA 12.6 under Windows; the Blackwell, A100, and H200 instances run torch 2.8.0 on CUDA 12.9, 2.10.0 on CUDA 12.8, and 2.8.0 on CUDA 12.9 respectively. `GEMMu18` is built at its revision `6bf409b`. The first call at a new problem shape additionally runs the dispatch probe and costs six to twelve times the steady-state call (16.6 ms against 1.41 ms at $n = 2048$ on Ada), and all timings exclude it. Accuracy is reported as correct bits, $-\log_2$ of the max-norm relative error against a reference accumulated in double-double arithmetic and cross-checked against exact rational arithmetic. The accuracy tables use $M = N = 128$: the error of both systems is driven by the inner dimension K , not by the output extent, and a spot check at $512 \times 4096 \times 512$ reproduces the pattern (native 49.4 bits, default 53.2, reduced 44.6).

`GEMMu18` is built from source and measured on the same hardware in every comparison: through its direct interface compiled natively on Windows for SM 8.9 and on Linux for SM 8.6, through its cuBLAS interposition hook on SM 8.6 and SM 9.0, and compiled in situ on the cloud instances. Where two measurement modes cover the same part they agree within three percent, and its accuracy under our double-double reference matches its published characterization, which cross-validates the harness.

The `fp64-emu default` configuration is self-sizing: fourteen moduli plus the two correction GEMMs through $K = 8192$, fifteen plus two beyond. The *reduced* configuration drops one modulus and the correction GEMMs, trading roughly eight bits of accuracy for speed. All numbers in this section are reproduced by scripts in the artifact [16].

5.2. Throughput against native FP64. Table 1 summarizes the five architectures at $n = 8192$, and Table 2 sweeps sizes on the part with the widest FP64-to-INT8 ratio. The ratio to native tracks each part’s throttling: $26\times$ on Blackwell and $16\times$ on consumer Ampere, whose native FP64 runs at $1/64$ rate, against $1.13\times$ on A100 and $1.02\times$ on H200, whose FP64 units are unthrottled and where emulation pays only at scale. Below $n \approx 128$ per-call cost exceeds native on every part; CUDA-graph replay moves this crossover to $n = 64$ on SM 8.9. Figure 2 plots the sweep on the three throttled parts.

TABLE 1. Throughput at $n = 8192$, square random-normal operands. *default* is the self-sizing configuration; *reduced* drops one modulus and the correction GEMMs.

part	SM	native	default	vs. native	reduced	vs. native
RTX PRO 6000 Blackwell	12.0	1.53 TF	39.89 TF	26.1×	47.91 TF	31.3×
RTX 3070 Ti Laptop	8.6	0.28 TF	4.45 TF	15.8×	5.25 TF	18.7×
RTX 6000 Ada	8.9	1.31 TF	14.92 TF	11.4×	18.66 TF	14.2×
A100-SXM4-80GB	8.0	17.58 TF	19.91 TF	1.13×	24.13 TF	1.37×
H200	9.0	58.18 TF	59.50 TF	1.02×	69.53 TF	1.20×

TABLE 2. RTX PRO 6000 Blackwell (Server Edition), SM 12.0, across sizes.

n	native	default	vs. native	reduced	vs. native
256	0.41 TF	0.60 TF	1.5×	0.62 TF	1.5×
1024	1.37 TF	11.40 TF	8.3×	12.44 TF	9.1×
2048	1.51 TF	21.76 TF	14.4×	25.69 TF	17.0×
4096	1.52 TF	32.40 TF	21.3×	38.55 TF	25.4×
8192	1.53 TF	39.89 TF	26.1×	47.91 TF	31.3×
16384	1.53 TF	39.95 TF	26.0×	48.01 TF	31.3×

TABLE 3. **GEMMu18** correct bits by modulus count, $K = 4096$, random-normal / spike.

moduli	14	15	16	17	18
bits	48.1 / 46.3	52.3 / 50.2	53.0 / 52.7	53.0 / 53.3	53.0 / 53.2

5.3. Comparison with GEMMu18. The comparison of central interest is against the scheme’s reference implementation, and its fairness rests on configuring both systems at their smallest FP64-equivalent settings. For **fp64-emu** that is the self-sizing default. For **GEMMu18** we establish it by measurement: Table 3 sweeps its modulus count under our double-double reference, and sixteen moduli reach the FP64-equivalent class at $K = 4096$. Across depths, sixteen moduli hold 52.2 to 53.3 bits through $K = 8192$ and measure 51.8 on the spike distribution at $K = 16384$, where seventeen restore 53.2. Every comparison therefore holds the GEMM count equal: **fp64-emu**’s fourteen moduli plus two correction GEMMs run against **GEMMu18** at sixteen moduli, and fifteen plus two against seventeen at $n = 16384$. **fp64-emu** measures 53.1 to 53.9 correct bits across the grid and **GEMMu18** measures 51.8 to 53.3, so the equal-count pairing gives **GEMMu18** the benefit of a slightly lower accuracy floor.

Table 4 reports the three throttled architectures, where **fp64-emu** is faster in every cell. The margin is largest at small sizes, $3.80\times$ at $n = 512$ on Ada, where the fused single-launch pipeline amortizes what **GEMMu18**’s multi-kernel structure pays per call; **GEMMu18**’s small- n throughput agrees across its Windows driver, Linux driver, and Linux hook measurements on SM 8.6, so this cost is a property of its pipeline rather than of any one platform. The margin narrows to 7 to 30 percent at the largest sizes, where both systems converge toward the throughput of the underlying INT8 GEMMs and the remaining separation comes from the byte-wide epilogue, the extraction, and the reconstruction.

Table 5 reports Hopper, where the picture inverts at scale: **fp64-emu** leads at $n = 2048$, the systems are within a few percent through $n = 8192$, and **GEMMu18** leads by 13 percent at $n = 16384$. Hopper pairs unthrottled FP64 with exceptionally strong cuBLAS INT8 kernels, and **GEMMu18**’s

TABLE 4. Equal-GEMM-count throughput in TFLOP/s on the three FP64-throttled architectures. Ratios are `fp64-emu` over `GEMMu18`. Dashes: $n = 16384$ exceeds the 8 GB consumer part; the two smallest Blackwell cells were not captured on the metered instance.

n	consumer Ampere (8.6)			Ada (8.9)			Blackwell (12.0)		
	ours	<code>GEMMu18</code>	ratio	ours	<code>GEMMu18</code>	ratio	ours	<code>GEMMu18</code>	ratio
512	0.94	0.64	1.47	2.85	0.75	3.80	–	–	–
1024	2.10	1.43	1.47	8.28	4.40	1.88	–	–	–
2048	3.58	2.27	1.58	12.71	9.46	1.34	21.76	13.14	1.66
4096	3.65	3.14	1.16	13.56	12.20	1.11	32.40	20.65	1.57
8192	4.45	3.80	1.17	14.92	13.78	1.08	39.89	30.78	1.30
16384	–	–	–	14.63	13.72	1.07	39.95	34.88	1.15

TABLE 5. Equal-GEMM-count throughput in TFLOP/s on H200 (Hopper, SM 9.0).

n	<code>fp64-emu</code>	<code>GEMMu18</code>	ratio
2048	22.63	19.95	1.13
4096	43.68	44.86	0.97
8192	59.50	58.61	1.02
16384	57.67	66.53	0.87

pipeline extracts more of that throughput at scale than ours does; both systems trail native FP64 below $n = 8192$ on this part.

At reduced precision, `fp64-emu` at thirteen GEMMs against `GEMMu18`’s thirteen-modulus fast mode, the throttled-part pattern repeats at smaller margins: $1.03\times$ to $2.92\times$ on consumer Ampere and Ada, and $1.01\times$ to $1.38\times$ on Blackwell, against $0.74\times$ to $0.97\times$ on Hopper. The one exception on the throttled parts is $n = 16384$ on Ada, where the self-sizing range rule fields fourteen moduli against thirteen and `GEMMu18` leads by 6 percent in proportion to the extra GEMM; pinning thirteen moduli at that depth restores parity, 17.28 against 17.60 TF, which localizes the deficit to the modulus count rather than to any per-GEMM inefficiency.

5.4. Component ablations. Three of the contributions can be isolated. *Sizing*: worst-case provisioning for a 53-bit guarantee at $K = 4096$ requires sixteen moduli (Section 3.1), eighteen GEMMs with the correction pair; the runtime pass delivers 53.1 to 54.1 bits from fourteen moduli, sixteen GEMMs, on all six distributions of Table 6, a saving of two GEMMs, narrowing to one at $K = 16384$. *Engine*: pinning the dispatch to one engine with everything else fixed separates the fused kernel from cuBLAS. On Ada in a single interleaved sweep, the fused engine delivers 7.34 against 5.30 TFLOP/s at $n = 1024$ and 12.52 against 11.39 at $n = 2048$, while the two agree within one percent at $n = 8192$ (15.77 against 15.61); on consumer Ampere the ordering inverts at large sizes, where the cuBLAS form leads by ten percent. No fixed rule captures both parts, which is what the measured dispatch replaces. *Reconstruction*: the register-resident grouped form of Section 3.3 is a component-level $1.6\times$ on SM 8.6 and $1.4\times$ on SM 8.9 over the per-modulus 128-bit recurrence it replaced.

5.5. Accuracy. Table 6 reports correct bits on six input distributions at two depths, measured on the Blackwell part; the reconstruction is exact integer arithmetic and carries no architecture dependence, and spot checks on the other parts agree within measurement spread. The distributions include per-element exponent jitter of eight and sixteen binades, shared column scalings spanning twenty and forty binades, and a 2^{30} spike element per row.

TABLE 6. Correct bits against a double-double reference, $M = N = 128$. Native, default, and reduced are measured on the Blackwell part; `GEMMu18` at sixteen moduli is measured through its hook on consumer Ampere, the reconstruction being architecture-independent. The default exceeds native FP64 DGEMM in every cell.

distribution	K	native	default	reduced	<code>GEMMu18</code>
randn	512	49.4	53.4	45.9	52.7
randn	4096	47.6	53.6	44.3	53.0
wide8	512	49.4	53.7	46.2	52.9
wide8	4096	47.6	53.7	45.1	52.5
wide16	512	49.4	53.3	46.8	53.2
wide16	4096	48.1	53.7	45.2	52.8
illcond	512	49.1	53.7	47.1	53.2
illcond	4096	47.7	53.3	45.4	52.6
illcond40	512	49.6	53.6	48.6	52.9
illcond40	4096	48.0	53.5	46.9	52.9
spike	512	48.7	54.0	45.7	53.3
spike	4096	47.8	53.2	43.3	52.7

TABLE 7. Batched throughput, RTX PRO 6000 Blackwell. The final column is the speedup of the batched call over looping the single call.

batch $\times$ n	native <code>bmm</code>	<code>fp64-emu bmm</code>	vs. native	vs. loop
64×256	1.55 TF	2.91 TF	$1.9\times$	$4.5\times$
32×512	1.57 TF	7.00 TF	$4.4\times$	$1.6\times$
16×1024	1.58 TF	13.92 TF	$8.8\times$	$1.1\times$
8×2048	1.58 TF	23.12 TF	$14.6\times$	$1.1\times$
4×4096	1.59 TF	32.71 TF	$20.5\times$	$1.1\times$

The mechanism behind the uniform advantage over native DGEMM is structural, not statistical: each emulated inner product is formed exactly in the residue domain and rounded once, where native DGEMM accumulates K roundings, so the default sits at the output-rounding floor while native drifts downward with depth. The exactness class is sharper still: on representable inputs the product is bit-identical to exact arithmetic, verified by equality against integer oracles across every dispatch path and, through the segmented pipeline, at $K = 200,000$.

5.6. Batched products. Iterative solvers and blocked factorizations, the workloads with the most to gain from a fast accurate product, issue many small GEMMs rather than one large one, and per-call overhead then dominates exactly where emulation’s raw advantage is thinnest. Table 7 measures the batched entry point against native `torch.bmm` and against looping the single-product call.

The pattern on SM 8.9 is the same, $1.6\times$ to $9.7\times$ native across the grid, and the loop amortization is architecture-independent: on A100 and H200, where native FP64 keeps the absolute lead at these sizes, batching remains $3.1\times$ and $4.0\times$ faster than looping at 64×256 . A batch of 64 products of dimension 256, a regime where the single-call emulation only breaks even against native FP64 on the throttled parts, runs at nearly twice native throughput batched.

5.7. End to end: a tiled Cholesky factorization. The batched entry point exists for blocked factorizations, so we close the evaluation with one. A tiled right-looking Cholesky factorization routes its trailing updates, batched tile products of uniform 512^3 GEMMs, through either `torch.bmm` or `emu.bmm`; the panel factorization and triangular solves stay native in both configurations, and both share identical gather and scatter code. On Ada at $n = 16384$ the update phase drops from 1.23 to 0.69 seconds and the whole factorization from 1.43 to 0.94 seconds, a $1.5\times$ end-to-end speedup; consumer Ampere at $n = 8192$ moves from 0.83 to 0.56 seconds. The non-update remainder differs slightly between the configurations, 0.20 against 0.25 seconds on Ada, reflecting allocator and cache state left by the different update engines rather than any difference in the shared code. The factorization residuals $\|A - LL^\top\|_F / \|A\|_F$ are equal between the two engines at 3 to 4×10^{-16} .

5.8. Energy. Board power sampled by `nvidia-smi` during sustained loops converts the throughput advantage into an energy one. On Ada at $n = 8192$, native FP64 draws 180 W for 840 ms per product, 151.6 J per GEMM, while the default configuration draws 297 W for 83 ms, 24.7 J per GEMM: the emulation pulls 65 percent more power and finishes ten times sooner, so it is $6.1\times$ more energy-efficient, 44.5 against 7.3 GFLOP per joule. The asymmetry in draw is itself informative: the throttled FP64 pipe cannot fill the machine and leaves the board underdrawn, while the emulation runs the tensor cores at the full power envelope and still wins on energy by finishing an order of magnitude sooner.

5.9. NVIDIA’s cuBLAS FP64 emulation. cuBLAS gained FP64 emulation on Blackwell-class parts in CUDA 13.0 Update 2 [10]. We linked cuBLAS 13.6 on consumer Ampere and on Ada and enabled the emulation strategy: output and timing remain bit-identical to native FP64 on both, so the feature does not engage below Blackwell, and on those architectures the residue schemes are the only accelerated FP64 paths. On the RTX PRO 6000 Blackwell, the one part both systems cover, NVIDIA reports up to a $13\times$ speedup over native FP64; `fp64-emu` measures $26.1\times$ on the same part at accuracy above native.

6. DISCUSSION AND LIMITATIONS

Where the remaining headroom is. At the largest sizes the comparison in Table 4 compresses toward the throughput of the underlying INT8 GEMMs, and there our engine and cuBLAS’s are already at parity: on SM 8.9 both sustain within one percent of each other under identical conditions. Profiling attributes the residual distance to the tensor-core issue ceiling of the `mma.sync` instruction class on SM 8.0 through 8.9, which vendor kernels share; the instruction sets that break that envelope, `wgmma` on Hopper and fifth-generation tensor core operations on Blackwell, are the natural next step for the fused engine on those parts. `GEMMu18`’s lead at scale on Hopper (Table 5) measures what better utilization of that class of hardware is worth.

Economics by part. On hardware with unthrottled FP64, emulation is a wash below $n = 8192$ and a modest win above (Table 1), and on Hopper the reference implementation leads ours at scale; the method’s constituency is the far larger population of parts where FP64 is throttled, and there `fp64-emu` leads at every size. On those parts the reduced configuration additionally offers a 45-bit operating point at up to $31\times$ native throughput.

Bounds. A single launch requires $K < 131072$, the segmented path extends this to 8.4 million, and batched calls require uniform member shapes with the product of plane count and batch at most 65535. The implementation targets NVIDIA architectures SM 8.0 and later, where the required instructions exist. Published binaries cover Linux x86-64, and the repository builds the identical source just-in-time elsewhere, including Windows.

Reproducibility. Every number in this paper is produced by scripts published with the artifact [16], including the double-double accuracy harness, the `GEMMu18` comparison drivers for all three

architectures, and the integer-oracle exactness suite. The kernel is a single CUDA file; the accuracy-relevant constants, the modulus table, the range rule, and the correction scales, appear in it exactly as described here.

7. CONCLUSION

Residue arithmetic turns the fastest units on a modern GPU into a double-precision engine, and the distance between a scheme and a system lies in the sizing, extraction, reconstruction, and dispatch that surround the modular products. By measuring the operands instead of provisioning for their worst case, canceling quantization error instead of narrowing to avoid it, reconstructing in registers with byte-parallel arithmetic, and choosing engines by measurement instead of architecture, `fp64-emu` runs double-precision matrix multiplication above native throughput by an order of magnitude on the FP64-throttled parts and above the scheme’s reference implementation at every size on those parts, while returning answers that are more accurate than native DGEMM and exact whenever exactness is representable. The measured frontier now sits at the issue ceiling of the current integer tensor-core instruction class; the scheme has room to move the moment the instruction set does.

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